

Milestone 4 Completion Report

Project Title: BankBot - An AI Chatbot for Banking FAQs (Brand: "Bank of Trust")

Milestone Title: 04 - Admin Panel Integration, Knowledge Base & System Auditing

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1. Executive Summary

This report documents the successful completion of Milestone 4, the final operational phase for the **"BankBot Nexus"** project. While Milestone 3 focused on the user-facing experience, Milestone 4 delivered a robust **Administrative Ecosystem** designed to make the chatbot sustainable and scalable in a real-world banking environment.

The primary objective was to transition the system from a static prototype to a **Self-Improving Intelligence**. This was achieved by implementing a comprehensive **Admin Nexus** that allows non-technical staff to monitor live traffic, update the AI's knowledge base dynamically, and audit interactions for compliance. The project is now a complete, end-to-end full-stack banking assistant solution.

2. Technical Implementation

To support the advanced operational requirements of Milestone 4, the system architecture underwent significant upgrades:

- **Hybrid Intelligence Engine:** The backend logic in `dialogue_manager.py` was refactored to implement a 3-tier decision tree. It now prioritizes **Deterministic Rules** (for secure transactions), falls back to the **NLU Model** (for intent recognition), and finally queries a **SQL-based Knowledge Base** (for static FAQs) if confidence scores are low.
- **Dynamic "Hot-Reloading":** A custom re-initialization routine was built into the Flask application (`app.py`). This allows the admin to trigger a model retraining process that updates the active Scikit-Learn vectorizers in memory without requiring a server restart, ensuring zero downtime.
- **Expanded Data Schema:** The SQLite database was migrated to a production-ready schema. New tables were introduced for `faqs` (to store dynamic Q&A) and `chat_logs` (to record comprehensive audit trails including intent predictions and confidence scores).

- **Role-Based Access Control (RBAC):** Secure session management was implemented to strictly separate "Retail User" sessions from "Corporate Admin" sessions, protecting sensitive administrative routes.

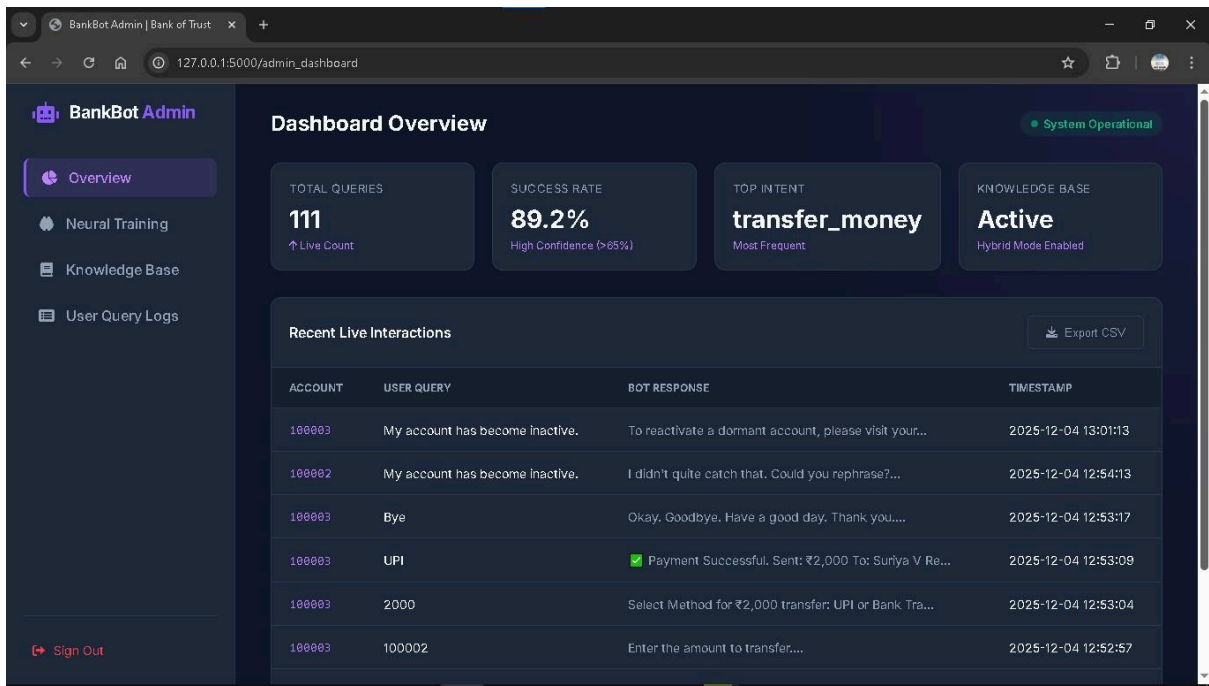
3. Key Deliverables

The following modules were developed, tested, and integrated during this final phase:

A. The Admin Nexus (Dashboard)

A centralized control room that provides a real-time health check of the system.

- **Features:** Displays live metrics on Total Queries, Model Success Rate (High vs. Low Confidence), and Top Detected Intents.
- **Impact:** Enables bank administrators to instantly gauge user satisfaction and identify areas where the bot is struggling.



(Caption: The Admin Overview tab showing real-time system analytics and interaction stats.)

B. Neural Training & Hot-Reloading

A dedicated interface for teaching the AI new concepts.

- **Features:** Admins can input new training examples (e.g., mapping slang like "plastic money" to `card_lost`) and click "Retrain Neural Network."
- **Impact:** Eliminates the dependency on developers for day-to-day model improvements.

C. Dynamic Knowledge Base (FAQ Manager)

A management tool for static banking queries.

- **Features:** A CRUD (Create, Read, Update, Delete) interface for adding Q&A pairs (e.g., "What is the SWIFT Code?").
- **Impact:** Acts as an immediate "patch" mechanism. If users start asking about a new policy, the bank can add the answer in seconds without waiting for a full model retrain.

D. Audit Logs & Compliance

A transparency tool for banking regulations.

- **Features:** A detailed log viewer showing every user message, the bot's response, the predicted intent, and the confidence score. Includes a "Export to CSV" function.
- **Impact:** Ensures full traceability of AI decisions, critical for resolving customer disputes and meeting audit requirements.

4. Challenges & Solutions

- **Challenge: Avoiding "Catastrophic Forgetting" during Retraining**
 - *Issue:* When retraining the model with new data, there was a risk of overfitting to recent examples and forgetting older, core banking intents.
 - *Solution:* Implemented a data aggregation pipeline that merges the new CSV inputs with the immutable "Core Training Set" before every training run. This ensures the model expands its knowledge without losing its foundation.
 - **Challenge: Handling Ambiguous Queries**
 - *Issue:* Users often asked questions that overlapped between the NLU model (e.g., "Balance") and the Knowledge Base (e.g., "Minimum Balance"), leading to conflicting responses.
 - *Solution:* Tuned the **Confidence Threshold** to 0.65. If the NLU prediction is below this score, the system automatically bypasses the intent engine and searches the Knowledge Base, ensuring specific factual answers take precedence over vague guesses.
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5. Conclusion

Milestone 4 marks the completion of the **BankBot Nexus** development lifecycle. By successfully integrating the Admin Panel, Dynamic Learning, and Audit capabilities, the project has evolved from a simple chatbot into a **comprehensive banking support platform**.

The system is now fully operational, capable of handling secure transactions, learning from new inputs, and providing deep operational insights. It stands ready for final evaluation and potential deployment as a proof-of-concept for modern FinTech automation.

Project Status: COMPLETE Next Steps: Final Code Cleanup, Documentation Handover, and Project Viva.